



HABIB
UNIVERSITY

Final Exam

Electromagnetic Theory

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Term: Fall 2025

Date: Dec 10, 2025

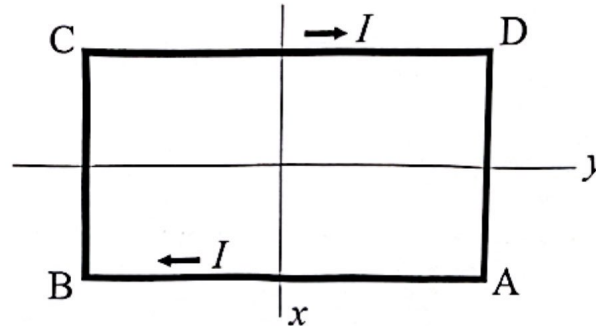
Points: 100

Instructions:

1. The final exam carries **30%** of the overall course grade.
2. The exam is open-notes and open-book.
3. Leave a left margin on each page. Use a pencil or pen and a ruler to draw the margin.
4. Bring your scientific calculator, extra pens, a ruler, water, and any glucose supplements you may need.
5. Cell phones must remain silent and remain inside your bags throughout the exam.
6. No pencil work is allowed except for drawing plots. Use a black pen for all written work.
7. Your work must be neat; show all necessary steps.
8. All diagrams must be clear, well-drawn, and properly labelled.
9. Begin each new question at the top of a fresh page.
10. Do not use the main answer booklet for rough work; request an extra sheet if needed.
11. To cancel any work, draw a box around it and place a single diagonal line through it. There is no need for excessive markings.

$$\int \frac{du}{(a^2 + u^2)^{3/2}} = \frac{u}{a^2 \sqrt{a^2 + u^2}} + c.$$
$$\int \frac{u du}{(a^2 + u^2)^{3/2}} = -\frac{1}{\sqrt{a^2 + u^2}} + c.$$

Q1: [40] A rectangular loop carries a current I [A] as shown in Figure 1 (below). Using Biot-Savart's law, calculate the magnetic field at the corner A of the loop. Width of the loop is $2b$ (along y -axis), and the height is $2a$ (along x -axis).



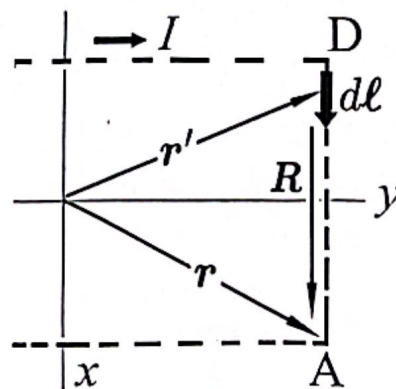
(a) Showing all steps (along edges AB, BC, CD, and DA), obtain $\mathbf{H}_{AB}$, $\mathbf{H}_{BC}$, $\mathbf{H}_{CD}$, and $\mathbf{H}_{DA}$ considering the field point is at A.

(b) Show that $\mathbf{H}_{\text{total}} = \mathbf{H}_{AB} + \mathbf{H}_{BC} + \mathbf{H}_{CD} + \mathbf{H}_{DA} = -\frac{I \sqrt{a^2 + b^2}}{8\pi ab} \mathbf{a}_z \quad \left[\frac{\text{A}}{\text{m}}\right]$

Use the following formula while specifying all involved variables in your calculations for each of the four edges:

$$d\mathbf{H} = \frac{I d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{4\pi \|\mathbf{r} - \mathbf{r}'\|^3}$$

Hint: An illustration is provided for edge DA as follows:

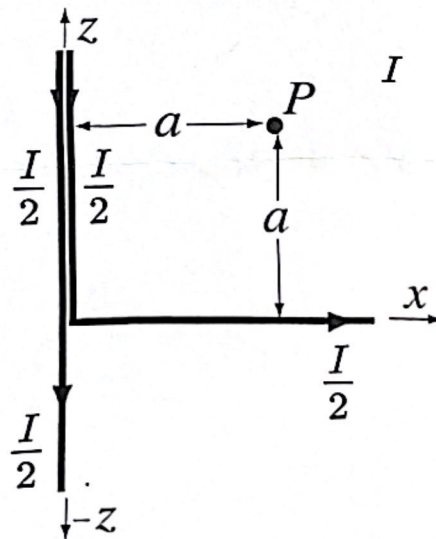
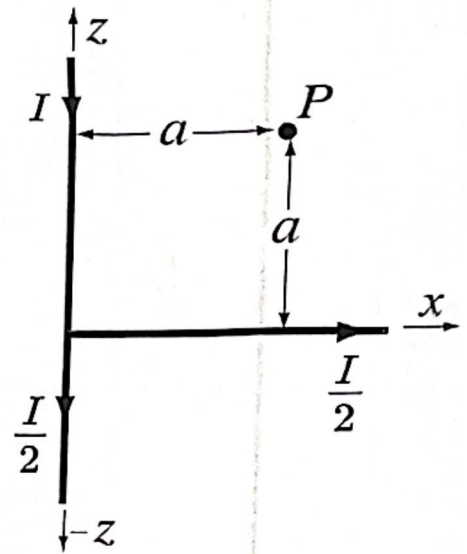


Provide similar illustrations for all edges in your work.

Q2: [30] An infinitely long wire carrying a current I [A] splits into two branches, each carrying $0.5 I$ [A] as shown in Figure 2. Determine the magnetic flux density at point $P(a, 0, a)$ using the Biot-Savart's law:

$$d\mathbf{B} = \mu_0 \frac{I d\boldsymbol{\ell} \times (\mathbf{r} - \mathbf{r}')}{4\pi \|\mathbf{r} - \mathbf{r}'\|^3}$$

Use this formula while specifying all involved variables. One may formulate this problem as two separate conductors (find a hint below).

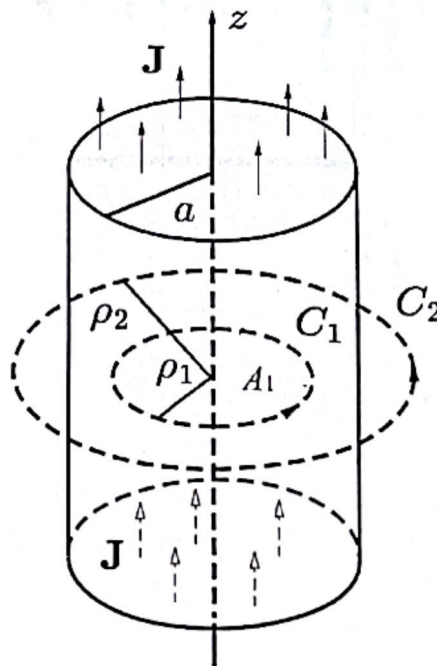


Q3: [30] Consider an infinitely long, straight cylindrical conductor with radius a , carrying a steady current I . The current is non-uniformly distributed such that the current density $\mathbf{J}$ is proportional to the radial distance ρ from the z -axis. Assume $\mathbf{J} = K(\rho^2 - 1) \mathbf{a}_z$ for $\rho \leq a$, where K is a positive constant. This is an obvious fact that $\mathbf{J} = \mathbf{0}$ for $\rho > a$ (outside the conductor).

- (a) [5] Given a and I , what is the value of K ?
- (b) [10] Using Ampere's law, determine the magnetic field intensity $\mathbf{H}$ both inside and outside the conductor.
- (c) [5] Plot the magnitude of $\mathbf{H}$ to show its variation as a function of ρ .
- (d) [10] Find the curl of $\mathbf{H}$ to establish that

$$\nabla \times \mathbf{H} = \mathbf{J}$$

is true everywhere, i.e., both inside and outside the cable.



When using Ampere's law, one can consider the circular Ampere contours C_1 and C_2 with radii ρ_1 and ρ_2 , respectively, where $0 < \rho_1 < a$ and $a < \rho_2 < \infty$, to calculate the magnetic field intensity both inside and outside the conductor.